

Spectra Scheduler

Smarter listening.
Under real receiver constraints.

Smart Scan strategy for Electronic Warfare

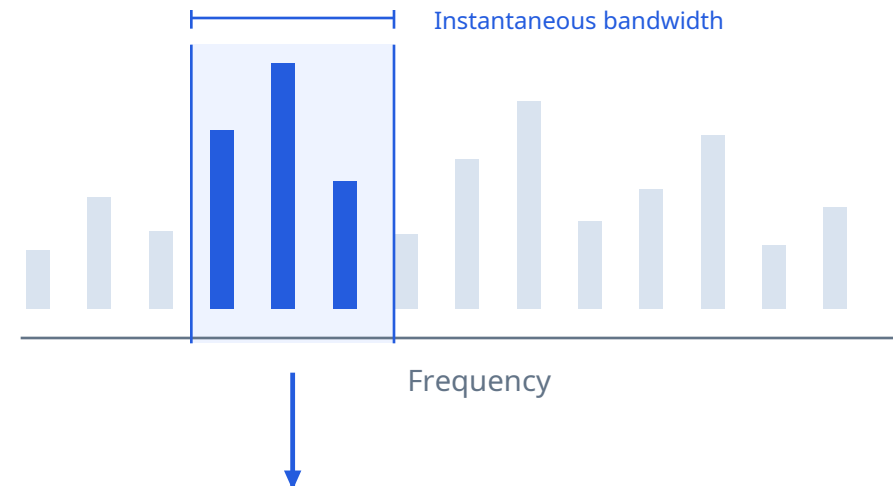
SIH26055 • DRDO • Software

Team name: _____ Team ID: _____

Theme: pending official confirmation

THE BANDWIDTH BOTTLENECK

WIDE SPECTRUM



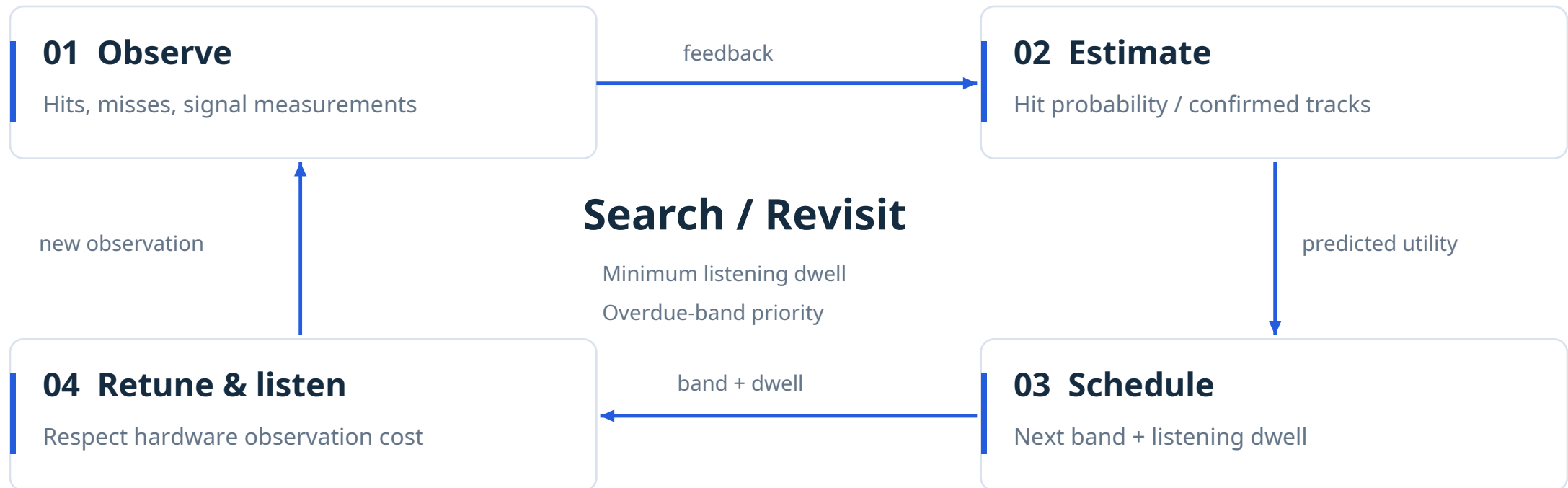
Limited-bandwidth receiver

Select band / dwell / revisit

**One observation window.
Many signals outside it.**

A closed loop, not a fixed sweep.

Proposed solution: adapt the observation schedule as evidence changes.

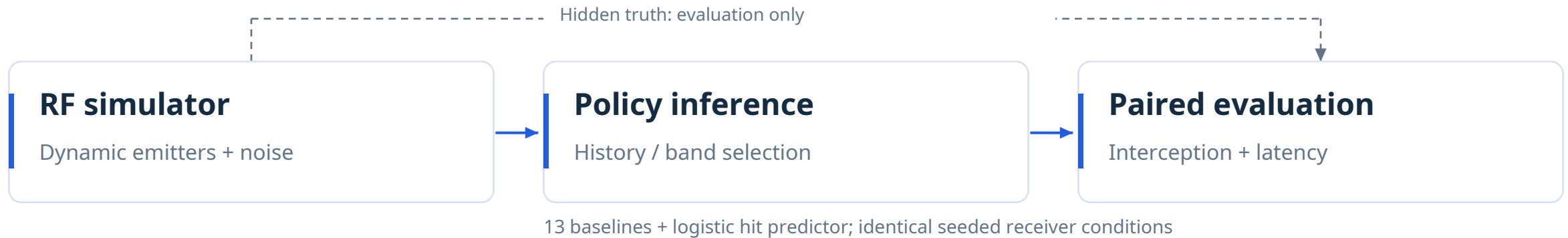


DISTINCTIVE DESIGN Observation-only inference • Drift response • Explicit receiver constraints

Two pipelines. One evaluation discipline.

Python / NumPy • h5py / HDF5 • scikit-learn • JSON model artifacts

A / CLOSED-LOOP SIMULATION



B / OFFLINE PULSE ASSOCIATION



Evidence first. Claims kept in scope.

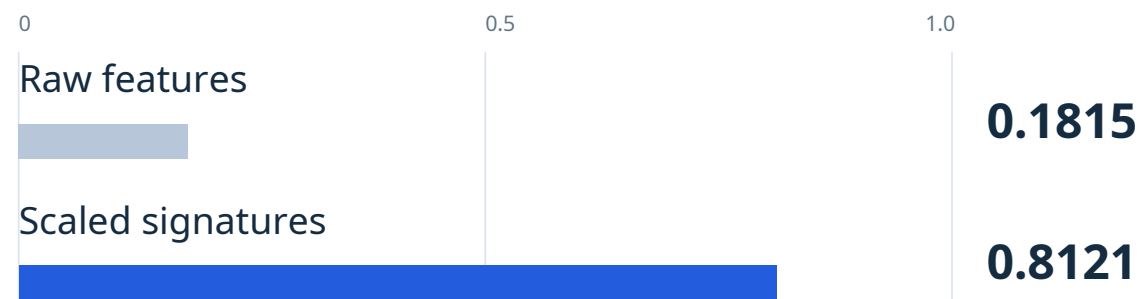
Implemented ingestion and association benchmarks; learned scheduling remains experimental.

233.2M

scan-training pulses validated

2,500 files • chunked HDF5 processing
Bounded samples; no full-dataset RAM load

ASSOCIATION / PAIRWISE F1



10 training files • 100k sampled pulses • not held-out

Incomplete observations

Truth-separated evaluation
Retuning excluded from targets

Scale & domain shift

Chunked I/O + bounded sampling
Unit calibration remains open

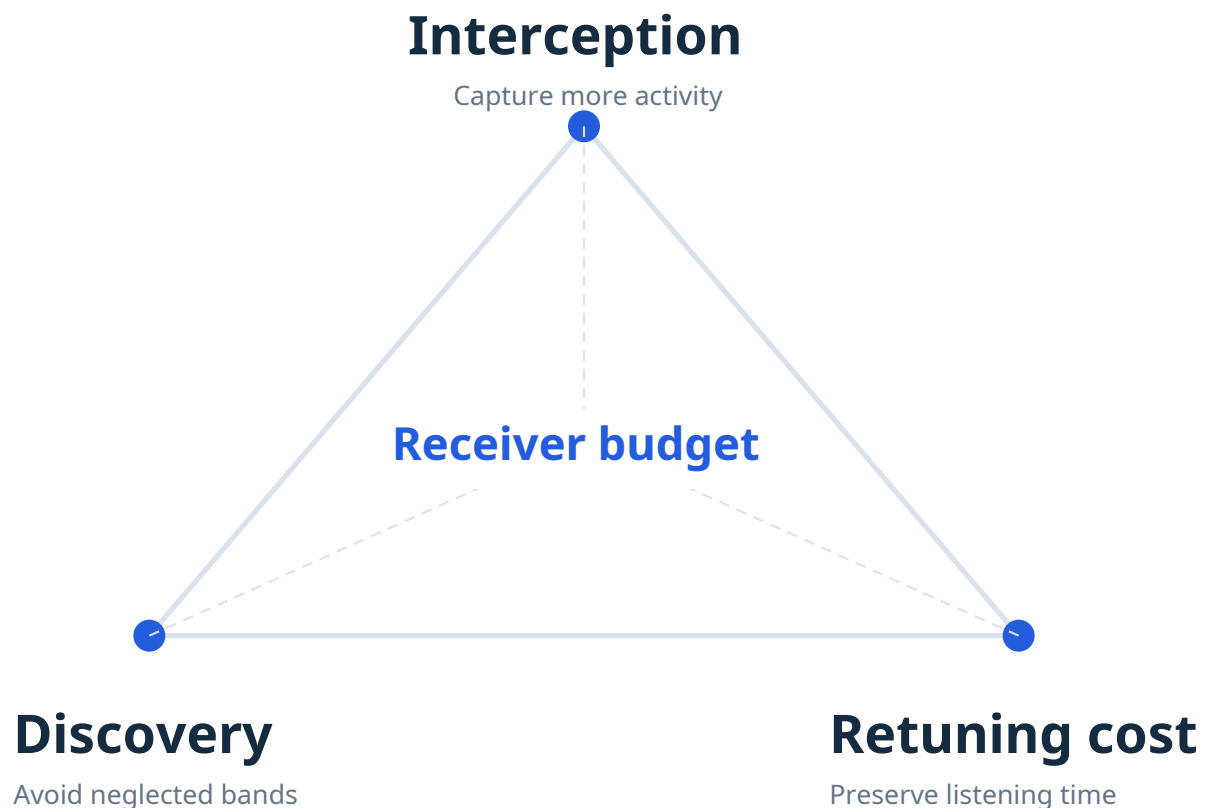
Unproven policy gain

Constant-model ablation
Existing defaults retained

VALIDATION GATE Learned policy remains experimental; constant-model ablation isolates the learning effect.

Optimise the trade-off. Not a single metric.

Intended users: receiver-algorithm researchers and spectrum-system evaluation teams.



Earlier detection

Measure first-detection and reacquisition delay.

Repeatable experimentation

Compare policies under identical seeded conditions.

Controlled compute demand

Stream data; profile memory and decision latency.

PLANNED DELIVERY Calibration bridge / broader validation / experiment API / dashboard

Grounded in research. Tested independently.

References inform the methods; published scores are not presented as project results.

- 01** **Benchmark foundation**  **The Turing Synthetic Radar Dataset**
Gunn et al. / arXiv:2602.03856

- 02** **Data & evaluation contract**  **TSRD dataset card + challenge implementation**
PDW schema, receiver modes, splits and scoring

- 03** **Representation-learning research**  **Radar Pulse Deinterleaving with Transformer**
Based Deep Metric Learning / arXiv:2503.13476

PROJECT EVIDENCE dataset-workflow.md / learning-workflow.md / reproducible JSON reports